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10 JUNE 1953
SUPERSEDING
Fed. Spec. U–I–496
2 October 1940

FEDERAL SPECIFICATION

IMPRESSION MATERIAL; HYDROCOLLOIDAL, AGAR TYPE, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Scope.—This specification covers dental impression materials containing a reversible, thermoplastic hydrocolloid (agar-agar type) as the gel-forming ingredient.

1.2 Type and grade. — This specification covers one type and one grade of impression material, hydrocolloidal.

2. APPLICABLE SPECIFICATIONS

2.1 There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Single copies of product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C. Prices of additional copies may be obtained from the GSA Regional Offices.)

3. REQUIREMENTS

3.1 Material.—The mixture of ingredients shall be combined in such proportions that the resulting material shall be uniform and suitable for taking impressions in the oral cavity or for reproducing dental models. Material may or may not contain fibers, as specified. Material shall not irritate oral tissues, and shall have a pleasant odor and flavor. The material shall impart a glossy surface to, and separate cleanly from, the hardened gypsum model. The manufacturer shall furnish a certificate stating that the material supplied does not contain poisonous ingredients in sufficient concentrations to be harmful to human beings when used as directed.

3.1.1 *Certification.* — Agar-type hydrocolloidal impression material shall have been tested at the National Bureau of Standards for the purpose of establishing a list of certified materials. Bidders shall attach to their bid a certificate and data showing that the brand of material they propose to furnish is the same in quality and brand as that appearing on the list of certified materials.

3.1.2 The material shall become sufficiently plastic when heated 5 minutes in a metal mixing syringe in boiling water or 10 minutes in the original container, to permit removal from the syringe or container and to permit adaption to the impression tray. After softening and during hardening of the material, the ingredients shall not segregate. The heated material shall be homogeneous, shall have a smooth surface, and shall be free from lumps or granules. The original container shall be such that it and the contained ma-

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terial can be boiled for 15 minutes without rupture of container.

3.2 Mold growth.—Material maintained 1 week at 20° to 25° C. shall not support mold growth. (See 4.4.3.)

3.3 Temperature of gel formation.—The temperature at which the plastic mass sets to a gel shall not be less than 35° C. (95° F.) nor more than 45° C. (113° F.). (See 4.4.4.)

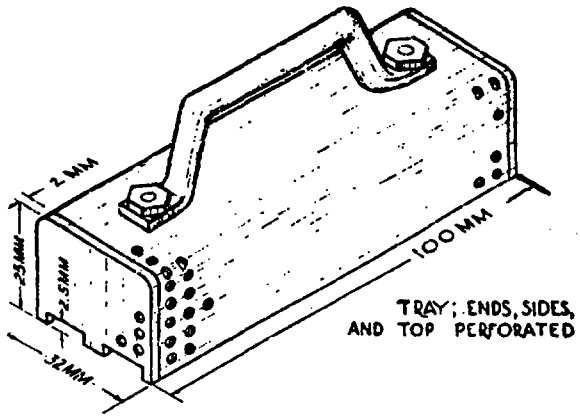
3.4 Integrity of impression.—Impression material shall not rupture during withdrawal from a model (see fig. 1) of which an impression has been taken. (See 4.4.5.)

3.5 Separation from model.—The impression material shall impart a smooth glossy surface to, and separate cleanly from, the hardened gypsum model produced from the impression taken. (See 3.4.)

3.6 Set.—The set shall be not more than 3.0 percent after 12 percent strain. (See 4.4.6.2.)

3.7 Strength.—The compression strength shall be not less than 28 pounds per square inch when tested as specified in 4.4.6.3.

3.8 Instructions.—Instructions for manipulation of the impression material shall accompany each package. (See 5.2.)



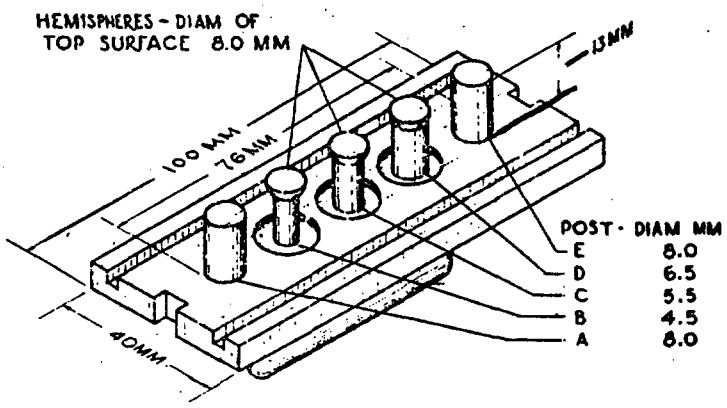


FIGURE 1.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—Unless otherwise specified, a lot shall consist of all agar-type impression material submitted by one manufacturer at the same time and manufactured by the same process.

4.2 Sampling.—At least one package of agar-type impression material shall be taken from each lot for government analysis.

4.3 Inspection.—Each sample shall be inspected for material, packaging, packing, and marking.

4.4 Physical tests.

4.4.1 *Standard testing conditions.* — All physical tests shall be made under uniform atmospheric conditions of 23° ± 1.1° C. (73.5° ± 2° F.) temperature and 50 ± 4 percent relative humidity. Equipment and materials shall be conditioned in the testing room for not less than 12 hours prior to making the tests.

4.4.2 *Preparation of test material.*—The impression material used for making all test specimens shall be prepared according to the instructions furnished with the package.

4.4.3 *Test for mold growth.*—Not less than 25 ml. of the test specimen shall be placed in the petri dish or similar container and cooled to 20° to 25° C. (68° to 77° F.). Mold obtained from stale bread or similar substance shall be used to inoculate the material by making three parallel lines through the material with a needle which shall have been previously drawn through the bread mold.

4.4.4 *Test for temperature of gel formation.*—Approximately 50 ml. of the material which has been prepared as specified in 4.4.2, shall be ejected into a small metal tray (fig. 2). As the material cools, a metal tube shall be pressed vertically into it until the floor of the metal tray is contacted. The tube shall then be immediately withdrawn. Near the critical point, trials shall be made every 0.5° C. The final trials shall be made in that portion of the material lying near the thermometer bulb. The temperature of the gel formation shall be the highest temperature at which the two concentric circles caused by the inside and outside surfaces of the indenting tube are clearly and sharply outlined and the material does not cling to the polished surfaces of the tube. The rate of cooling of the specimen from 50° C. (122° F.) to the temperature of gel formation shall be 1.5° ± 0.5° C. (2.7° ± 1.2° F.) per minute. An average of three tests on the temperature of the gel formation shall be reported to the nearest degree centigrade. When the determined number falls midway between two numbers, the even number shall be recorded. (See 6.6.*a*.)

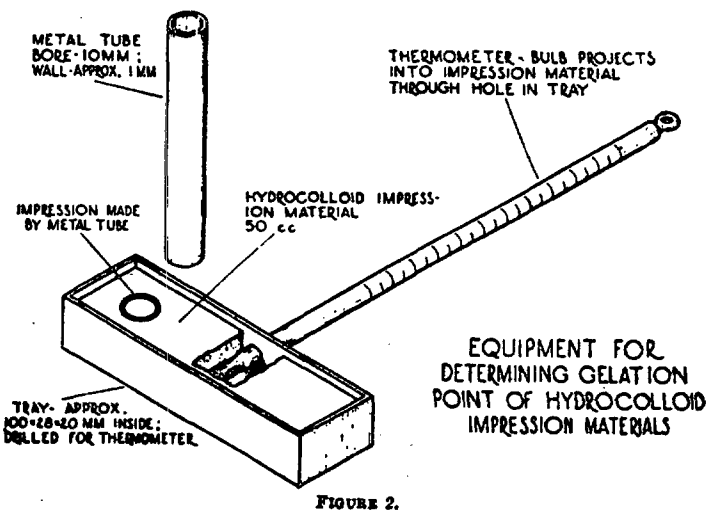


FIGURE 2.

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4.4.5 *Test for integrity of impression.*—An impression of the metal mold shown in figure 1 shall be taken in a tray with the impression material prepared as specified in 4.4.2. The impression shall be chilled in water at 10° ± 3° C. (50° ± 5° F.), for 15 minutes. The impression and model shall then be taken from the water and the impression pulled from the metal model. A gypsum model shall be immediately prepared in the impression by gently vibrating the unset gypsum mixture into it and allowing it to set for 30 minutes before removal.

4.4.6 *Set and compressive strength.*—The average values obtained on five specimens in the following tests shall be reported for set and compressive strength.

4.4.6.1 *Preparation of test specimens.*—A cylindrical ring of suitable diameter to confine the hydrocolloid material shall be placed on a flat glass plate and shall be slightly more than one-half filled with the impression material prepared as specified in 4.4.2. A metal mold 0.5-inch inside diameter, 1-inch outside diameter, and 0.75-inch high (see fig. 3) shall be placed inside the ring immediately and forced into the material until the mold comes in complete contact with the glass plate and the material has exuded on top of the mold. A flat plate shall then be pressed on top of the mold to remove the excess material. One-half hour later, the specimens shall be made flat and parallel. If necessary, the mold containing the specimen may be drawn across No. 00 abrasive paper. The specimens shall be removed from mold and placed in an atmosphere of 100-percent relative humidity at 23° ± 1° C. (73.4° ± 1.8° F.) for approximately ½ hour.

4.4.6.2 *Test for set.*—A test specimen (see 4.4.6.1) shall be placed in a suitable instrument (as outlined below) and measured for set.

4.4.6.2.1 The instrument used for measuring the set shall consist essentially of a dial gage, sensitive to 0.001 inch, mounted to a steady base and equipped with a screw, positioned in such a manner that sufficient pressure can be applied to the specimen to produce the required amount of strain (see fig. 3).

4.4.6.2.2 Place lightweight plate on top of specimen and allow dial gage to contact plate. Weight of plate and pressure of spring in dial gage shall not be more than 50 grams. Let stand for 30 seconds and read dial gage. Record as reading "A".

4.4.6.2.3 From reading "A" subtract 0.090 inch, which is 12 per cent of specimen length, and lower the dial gage by means of the screw until it registers the new figure. This shall be done 30 seconds after reading "A".

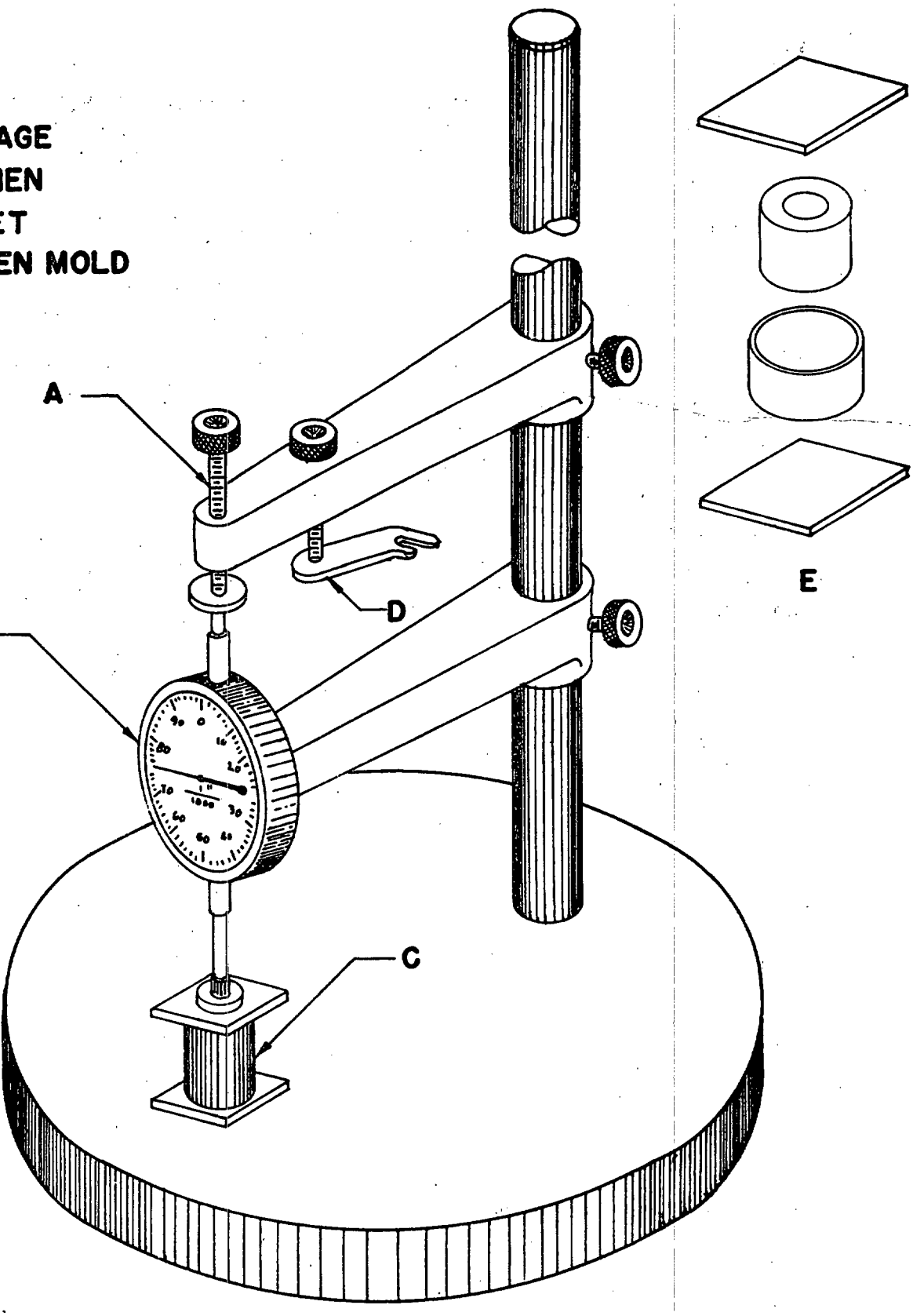
4.4.6.2.4 Maintain under pressure for 1 minute, then release all pressure on the specimen for 1 minute. Again lower the dial gage onto the specimen as indicated in 4.4.6.2.2, and after 30 seconds, take a second reading. Record as reading "B".

4.4.6.2.5 The difference between reading "A" and reading "B", divided by the original length of the specimen times 100 shall be recorded as the percent set at 12-percent strain. For calculating the set and strain, the length of the specimen shall be taken to be the length of the mold from which it is made.

4.4.6.3 *Test for compressive strength.*—The same specimen used in testing for set shall be immediately placed in a suitable machine, accurate to 0.1 pound, and tested for compressive strength. A piece of bond paper shall be placed under and over the specimen in the machine. The load shall be applied to the specimen slowly and continuously at a uniform head speed of 0.20 ± 0.05 inch per minute until the specimen fractures. The maximum load obtained before fracture shall be recorded to the nearest 0.1 pound. The average maximum load shall be divided by the cross section area of the mold and reported in pounds per square inch. For calcu-

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A- SCREW
B- DIAL GAGE
C- SPECIMEN
D- RATCHET
E- SPECIMEN MOLD



INSTRUMENT FOR MEASURING PERCENT OF SET

FIGURE 3

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lating the compressive strength, the cross-section area of the specimen shall be considered to be the same as that of the mold from which it is made.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, the material shall be supplied in sealed, airtight, collapsible tubes, which shall be made of substances that shall not have a chemical affinity for, nor contaminate the impression material, and which shall be sufficiently flexible to permit extrusion of the heated material. If a metal tube is used it shall have a clip securely attached. A package shall consist of 12 tubes packed in a suitable cardboard carton.

5.2 Instructions. — Adequate and accurate instructions for manipulation shall accompany each package. These instructions shall include:

1. Time and temperature requirements for softening the material in the original container and in a mixing syringe.
2. The temperature of the material when inserted in the mouth.
3. The technic for chilling the impression in the mouth.
4. The treatment of the impression between the interval of withdrawal from the mouth and preparation of the gypsum model.

5.3 Packing. — Unless otherwise specified, impression material shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.4 Marking.

5.4.1 *Packages.* — Unless otherwise specified, packages shall bear the regular trade label, stating the contents, name of the manufacturer or contractor, net weight of contents; shall be marked with a serial number or combination of letters and numbers which shall refer to the manufacturer's records for the particular lot; the date of manufacture (year and month) shall be indicated on the package as a separate item or as a part of the serial number.

5.4.2 *Shipping containers.* — Unless otherwise specified, shipping containers shall be marked with the name and type of material as defined by the contract or order under which the contract is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Intended use. — The material covered by this specification is intended as a reversible, thermoplastic hydrocolloidal impression material (agar-agar type). For nonreversible hydrocolloidal impression material, refer to Federal Specification U–I–498 for Impression Material, Hydrocolloidal, Alginate Type, Dental.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein. (See 3.1, 4.1, 5.1, 5.3, 5.4, 6.3, and 7.2.6.)

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 This specification covers the following item appearing in the Armed Services Catalog of Medical Matériel:

5–362–450—Hydrocolloid, Agar Type, 12s.

6.5 This specification does not include all types, classes, grades, etc., of the commodity

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indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally procured for Federal Government requirements.

6.6 Reference:

a. Journal, American Dental Association, Volume 27, Page 373, March 1940.

b. Journal of Dental Research, Volume 28, Page 573, December 1949.

6.7 Transportation description.—The proper transportation description for this item is:

Dental impression compound. (Value not to exceed 50 cents per pound.)
Carload weight 30,000 pounds.
Truckload weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specification:

NN–B–601—Boxes, Wood-Cleated-Plywood, for Domestic Shipment.

NN–B–621—Boxes; Wood, Nailed and Lock-Corner.

NN–B–631—Boxes; Wood, Wirebound (for Domestic Shipment).

Military Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

MIL–B–107—Boxes, Wood, Wirebound (Overseas Type).

JAN–P–120—Packaging and Packing for Overseas Shipment—Cartons, Folding, Paperboard.

JAN–P–133—Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.

MIL–L–10547—Liners, Case, Waterproof.

Navy Department Specification:

General Specifications for Inspection of Material.[1]

Military Standard:

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Unit packaging.*—Sufficient impression material to take a full size upper or lower impression (approximately 2 oz. net weight), shall be supplied in a sealed, airtight, collapsible tube, which shall not have a chemical affinity for, nor contaminate the impression material, and which shall be sufficiently flexible to permit extrusion of the heated material. Lower end of metal tube, if used, shall be secured by means of a standard fold and clip, or quadruple fold and corrugated seal, or shall be fused by induction welding to form a hermetic seal without the use of a fluxing agent. Twelve tubes of im-

[1] Applicable to Navy purchases only.

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pression material, as specified, constitute one unit. Each unit shall be packaged in a carton of appropriate size conforming to the requirements of Military Specification JAN–P–120 or JAN–P–133. Carton shall be fitted with half-slotted dividers so that each tube is contained in an individual cell. Each carton shall be adequately secured to prevent accidental opening.

7.2.3 *Packing.*

7.2.3.1 *For domestic shipment.* — Twenty-four unit cartons (24 units) shall be packed in a shipping container conforming to the requirements of Federal Specification NN–B–601, or in a shipping container designed for type 2 load, and conforming to the requirements of Federal Specification NN–B–621 or NN–B–631. Boxes shall be strapped when required by appendix to applicable specification.

7.2.3.2 *For overseas shipment.* — Twenty-four unit cartons (24 units) shall be packed in a shipping container designed for type 2 load, and conforming to the requirements of Military Specification JAN–P–105, JAN–P–106, or MIL–B–107. Closure and strapping shall conform to the requirements of the appendix to applicable box specification.

7.2.3.2.1 *Case liner.* — Each box shall be lined with a waterproof case liner conforming to the requirements of Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

7.2.3.3 *Packing variation permitted.* — If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in a shipping container of suitable size and design and so constructed as to insure safe delivery to the specified destination.

7.2.4 *Marking.*

7.2.4.1 *Packages.*—Each tube shall be commercially identified. Each unit carton shall be marked with stock number, nomenclature, unit, quantity, date of manufacture (month, year), lot (control) number, and name of the contractor.

7.2.4.2 *Shipping container.* — Each shipping container shall be marked in accordance with Military Standard MIL–STD–129.

7.2.5 *Inspection procedures (Navy only).* —The general inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

7.2.6 *Ordering data.* — Procurement documents should specify whether the subject commodity is to be packed for domestic or overseas shipment. (See 7.2.3.)

NAVY INTEREST: Md

ARMY INTEREST: M

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

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